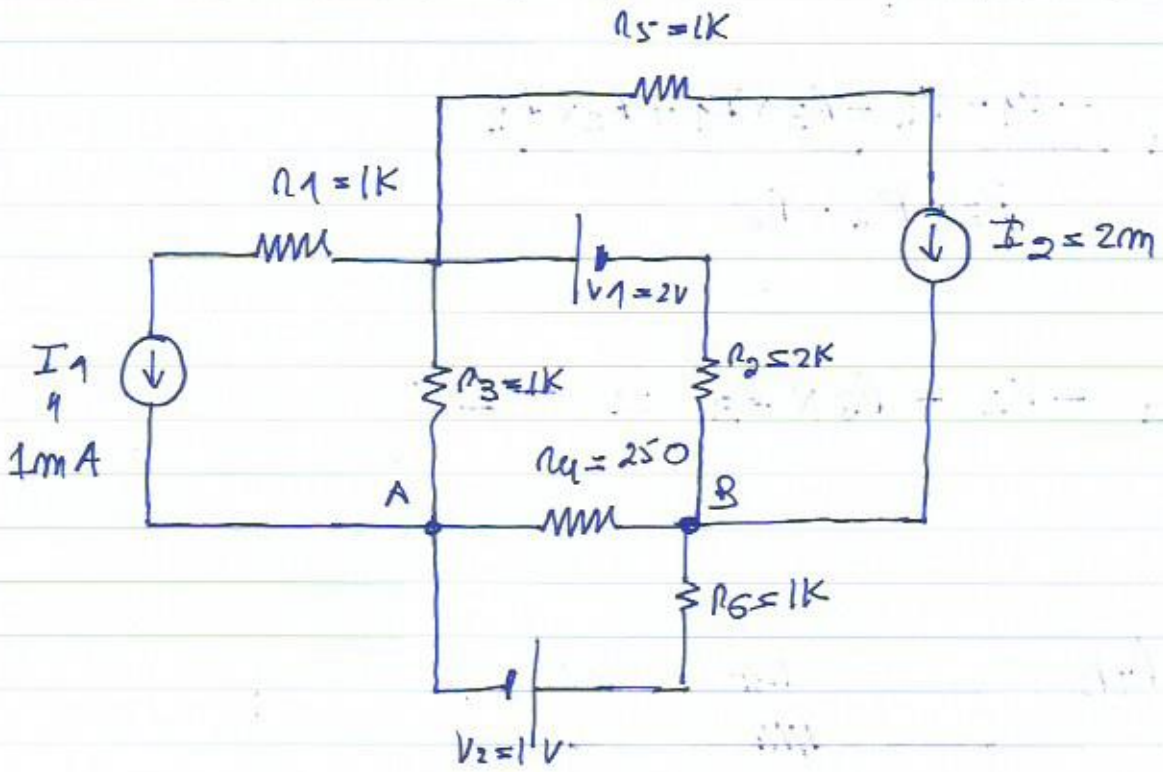
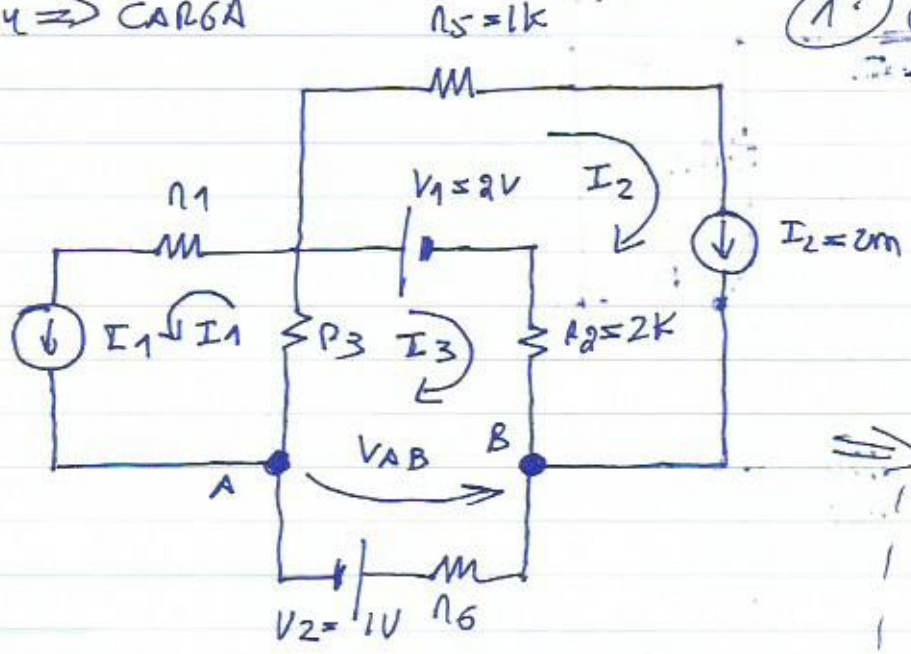
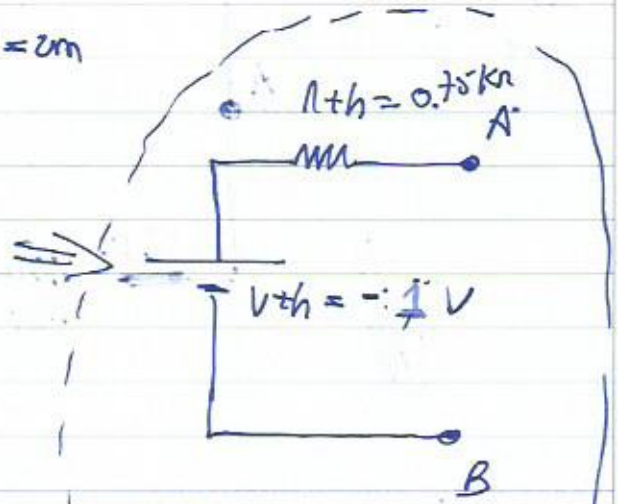
15)



R4 ⇒ CARGA



1° CÁLCULO Vth



EQUIVALENTE THEVENIN

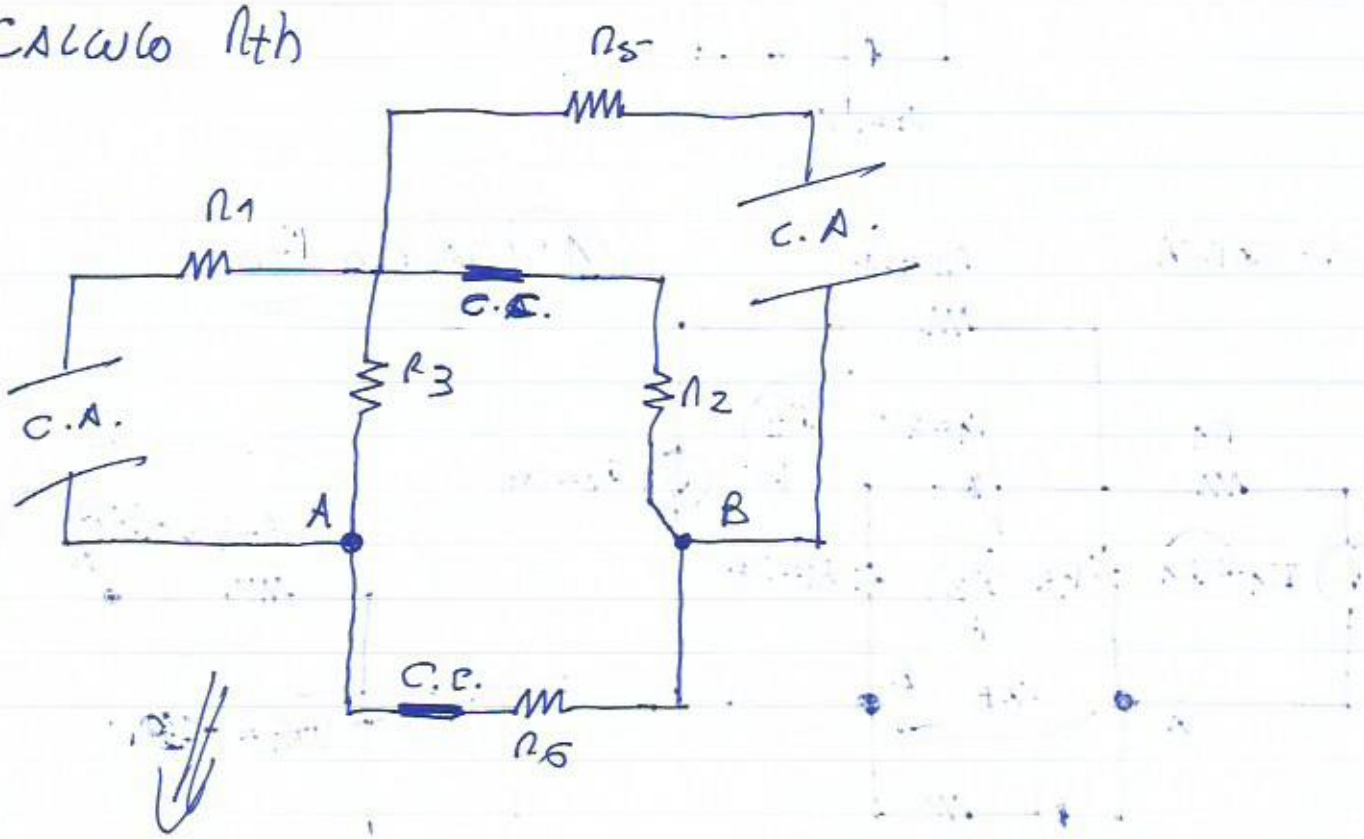
- $V_{AB} = -V_2 - R_6 I_3$
- $R_3 (I_3 + I_1) + V_1 + R_2 (I_3 - I_2) + R_6 \cdot I_3 + V_2 = 0$
- $I_1 = 1m$
- $I_2 = 2m$

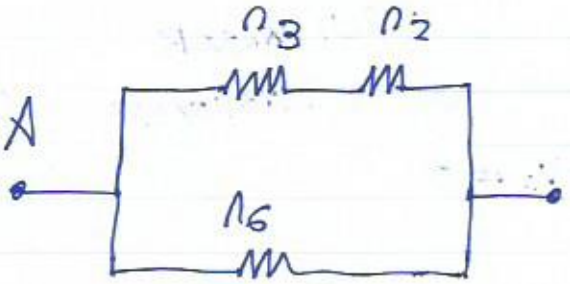
$$I_3 \times (R_3 + R_2 + R_6) = -V_1 - V_2 - R_3 \times I_1 + R_2 \times I_2$$

$$I_3 = \frac{-V_1 - V_2 - R_3 \times I_1 + R_2 \times I_2}{R_3 + R_2 + R_6}$$

$$V_{th} = -V_2 - R_6 \times I_3 = -1\,V$$

Calcolo R_{th}





$$R_{th} = (R_3 + R_2) /\!/ R_6 = 75\,\Omega$$

Nome

N.º Aluno

Curso

Ano Letivo

Data da Avaliação

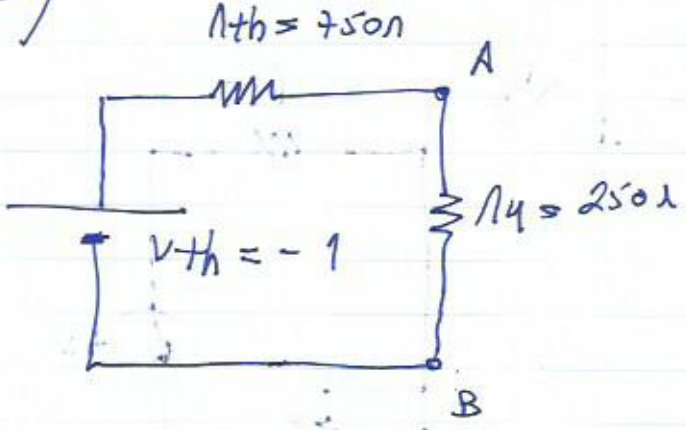
Prova Escrita de:

N.º Folhas

Época:

20

b)

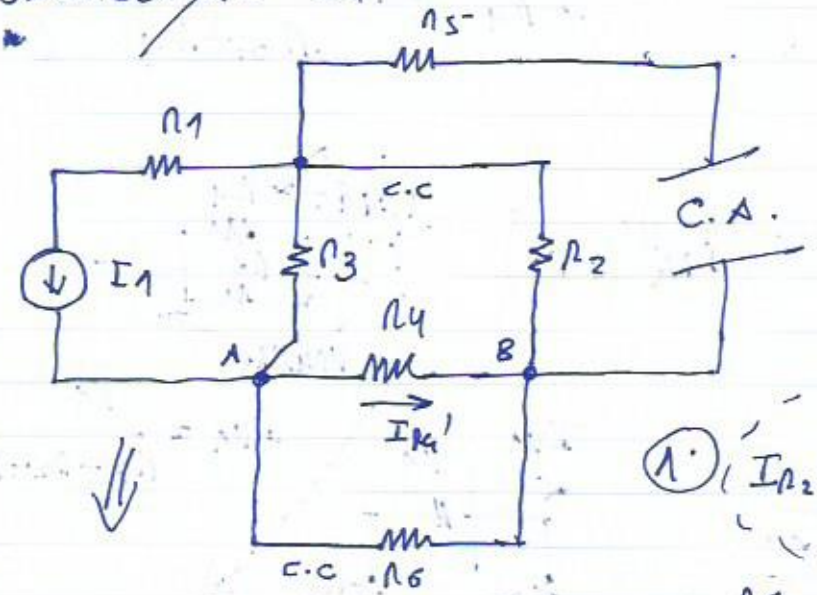


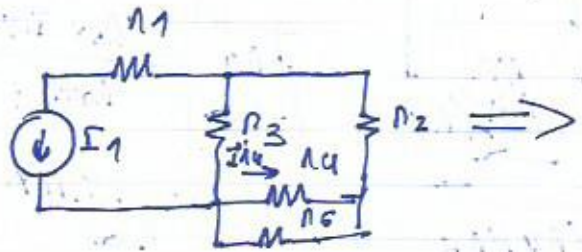
$$I_{R4} = \frac{V_{th}}{R_{th} + R_4} = \frac{-1}{750+250} = -1\,mA$$

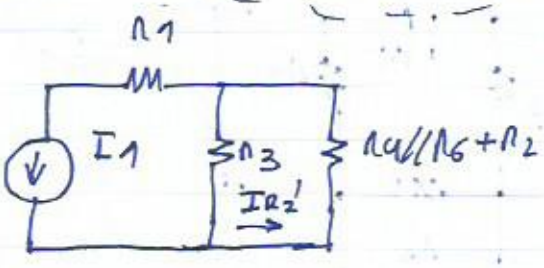
c) TEOREMA DA SOBREPOSIÇÃO.

I_{R4}' (→)

(1) CONTRIBUIÇÃO I_1.

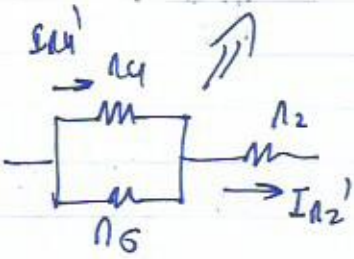




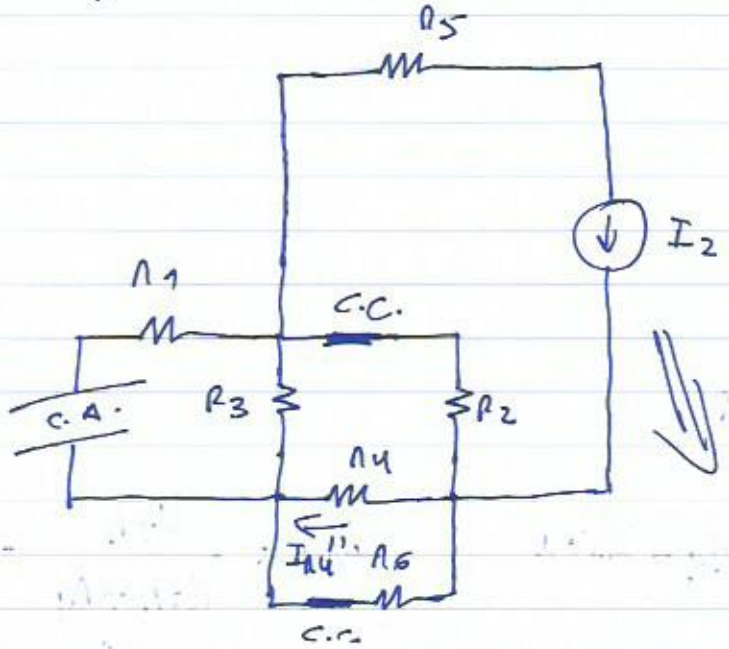


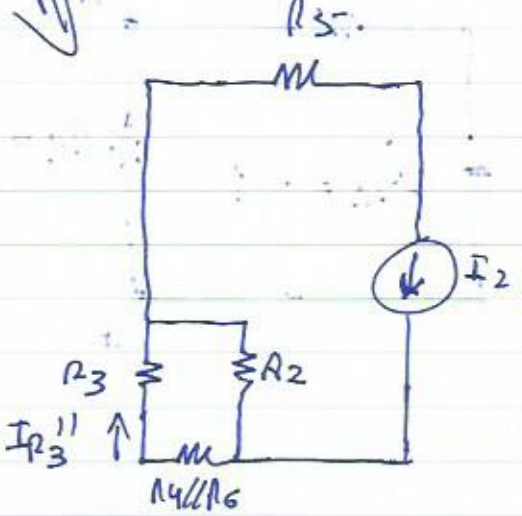
(1.) $I_{R2}' = \frac{R_3 \times I_1}{(R_4 // R_6 + R_2) + R_3}$

(2.) $I_{R4}' = \frac{R_6 \times I_{R2}'}{R_4 + R_6}$

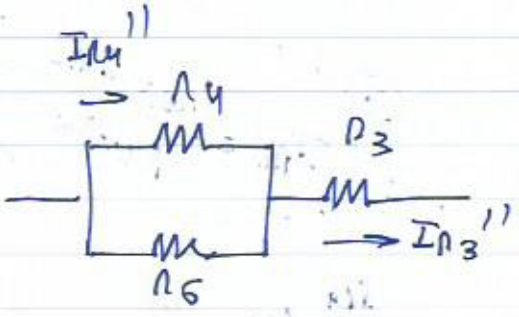


(II) contribuye I_2 I_{R4}'' (←)



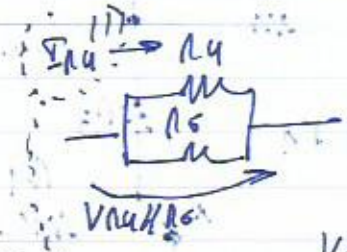


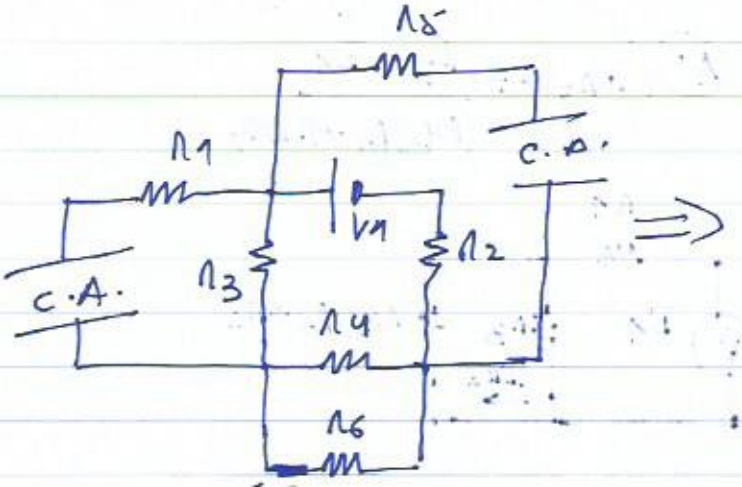
$$I_{R_3}'' = \frac{R_2 \times I_2}{R_2 + (R_3 + R_4 // R_6)}$$



$$I_{R_4}'' = \frac{R_6 \times I_{R_3}''}{R_6 + R_4} = 1mA \quad (\leftarrow)$$

(III) contribuye V_1 (I_{R_4}''' →)





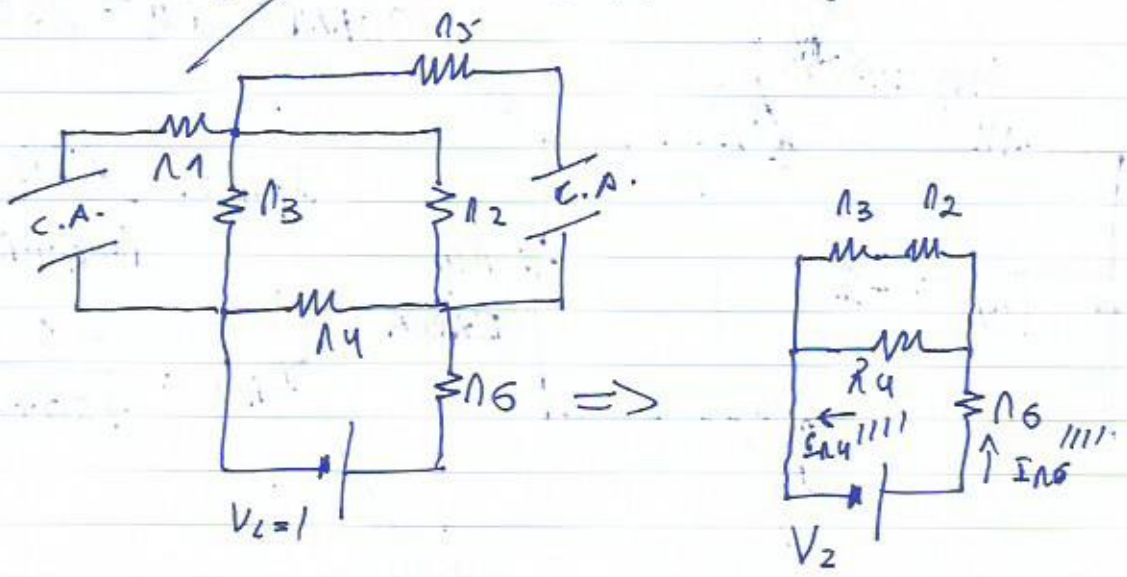


$$I_{V_1} = \frac{V_1}{R_3 + R_2 + R_4 // R_6}$$

$$I_{R_4}''' = \frac{R_6 \times I_{V_1}}{R_6 + R_4} \quad (\rightarrow)$$

$$= 0.5mA \quad (\rightarrow)$$

(IV) contribuição V_2 (I_{R4}'''' ←)



$$I_{R6}'''' = \frac{V_2}{R_6 + R_4 // (R_3 + R_2)}$$

$$I_{R4}''' = \frac{(R_3 + R_2) \times I_{R6}''''}{(R_3 + R_2) + R_4} = 0{,}75\ mA \quad (\leftarrow)$$

CONTRIBUIÇÃO I_1 ⇒ $I_{R4}' = 0{,}25\ mA$ (→)

CONTRIBUIÇÃO I_2 ⇒ $I_{R4}'' = 1\ mA$ (←)

CONTRIBUIÇÃO V_1 ⇒ $I_{R4}''' = 0{,}5\ mA$ (→)

CONTRIBUIÇÃO V_2 ⇒ $I_{R4}'''' = 0{,}75\ mA$ (←)

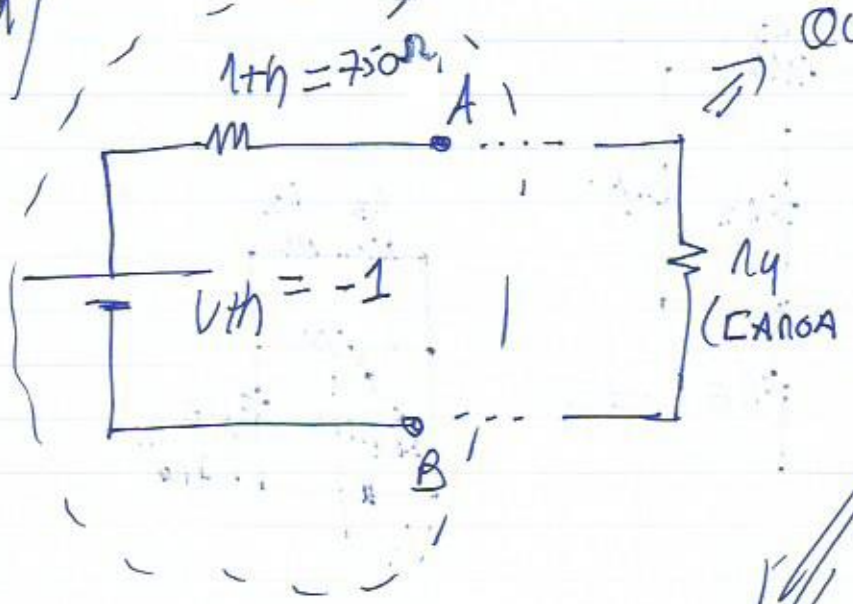
⇓

$$I_{R4} = I_{R4}' - I_{R4}'' + I_{R4}''' - I_{R4}''''$$

$$(\rightarrow) = 0{,}25m - 1m + 0{,}5m - 0{,}75m = 0{,}75m - 1{,}75m$$

$I_{R4} = 1\ m\ (\leftarrow)$

d) ALINEA (A)



QUANDO $R_L = 250\,\Omega$

$$I_{RL} = \frac{V_{th}}{R_{th} + R_L}$$

$$I_{RL} = \frac{-1}{750 + 250} = -1\,mA$$

Queremos $R_L \Rightarrow I_{RL_{NOVO}} = \frac{I_{RL}}{4} = -0{,}25\,mA$

$$-V_{th} + R_{th}\, I_{RL_{NOVO}} + R_{L_{NOVO}}\, I_{RL_{NOVO}} = 0$$

$$R_{L\,NOVO} = \frac{V_{th} - R_{th} \times I_{RL\,NOVO}}{I_{RL\,NOVO}}$$

$$R_{L_{NOVO}} = \frac{-1 - 750 \times -0{,}25\,m}{-0{,}25\,m}$$

$$R_{L\,NOVO} = \frac{-1 + 0{,}1875}{-0{,}25} \times 1000$$

$$R_{L_{NOVO}} = \frac{0{,}8125}{0{,}25} \times 1000 = \underline{\underline{3250\,\Omega}}$$